# Fatima Shahid

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## ABOUT ME

Empathetic and highly organized professional with a strong academic background and over five years of dedicated teaching and mentoring experience. Adept at breaking down complex information, managing rigorous administrative processes, and guiding students toward academic success. Combining a detail-oriented, analytical mindset with exceptional interpersonal skills, I am passionate about providing tailored, strategic advice to help students and their families make informed, confident decisions about their educational and career pathways.

## EDUCATION AND TRAINING

12/09/2022 – 30/06/2026 Islamabad, Pakistan

**BS PHYSICS** Pakistan Institue of Engineering and Applied Sciences

Main Subjects: Machine Learning, Quantum Mechanics, Quantum Optics, Quanum Computing, Computational Phyiscs, Probability and Statistics, Linear Algebra, Numerical Methods, Mathematical Modelling, etc.

**Field of study** Physics , Natural sciences, mathematics and statistics not further defined |

**Thesis** Fluorescence-Based Detection and Tracking of Trapped Particles in an Optical Tweezer System

## WORK EXPERIENCE

**EDUCATIONAL MENTOR & TUTOR** – 2017 – Current – FAISALABAD, LAHORE, JHELUM, PAKISTAN

- Provided comprehensive academic tutoring and mentorship to students across multiple educational levels for over five years.
- Taught mathematics, biology, and English to secondary school students, adapting teaching methodologies to suit diverse learning styles and individual academic needs.
- Designed and delivered targeted lessons for high school students in physics and chemistry, simplifying complex scientific concepts to improve student comprehension and confidence.
- Initiated and led a free tutoring program for underprivileged children in the local community, independently sourcing educational materials and dedicating weekly hours to teaching foundational literacy and numeracy.

**DIGITAL MARKETING & SALES INTERN – AMERICOS INDUSTRIES INC** – 02/06/2025 – 16/06/2025 – LAHORE, PAKISTAN

- Gained practical, hands-on experience in sales strategies, digital marketing, and direct client engagement.
- Contributed to campaign planning and content creation, successfully executing effective social media growth and audience interaction strategies.
- Recognized by management for contributing innovative ideas that enhanced the company's online presence and improved communication with the target audience.

**INTERN – PAKISTAN INSTITUTE OF ENGINEERING AND APPLIED SCIENCES** – 30/06/2025 – 08/08/2025 – ISLAMABAD, PAKISTAN

Supervisor: Dr. Sadiq Nawaz Khan, Associate Professor
Subject: Quantum Optics, Quantum Sensing, Optics

- Worked on advanced experiments in quantum optics, including laser alignment, beam manipulation, and interferometry.
- Gained hands-on experience with optical components such as beam splitters, polarizers, and wave plates.
- Assisted in data collection and analysis for studies involving light-matter interaction and photonic systems.

**ML INTERN – PAKISTAN INSTITUE OF ENGINEERING AND APPLIED SCIENCES (PIEAS)** – 01/07/2024 – 06/09/2024 – ISLAMABAD, PAKISTAN

Supervisor: Dr. Muhammad Asim Rasheed, Professor

Subjects: Machine Learning, Python, DFT

- Worked on predicting the volatility and boiling points of organic compounds using the Clausius-Clapeyron equation and machine learning models.

## VOLUNTEERING

08/09/2025 – CURRENT Islamabad

### Hostel Secretary

**Liaison & Advocacy:** Acted as the primary bridge between residents and administration, holding frequent meetings with authorities to mediate disputes, resolve grievances, and improve living conditions.
**Facility Management:** Oversee daily hostel operations, ensuring high standards of cleanliness and coordinating prompt repairs for infrastructure issues such as heating and room maintenance.
**Policy Development:** Formulate, propose, and implement new hostel rules and regulations to foster a disciplined and supportive residential community.

Lahore

### Shaukat Khanum Cancer Hospital

Served as a dedicated volunteer at a cancer hospital, focusing on patient advocacy and public health education. Successfully organized and executed awareness events to educate the community on early detection and prevention. Provided compassionate bedside support and logistical assistance to vulnerable patients, ensuring their comfort and well-being during treatment.

### Al-Khidmat Organization

CURRENT

### Community Volunteer

Initiated and led a free tutoring program for underprivileged children in the local community. Also sourced educational materials and dedicated hours weekly to teaching literacy and numeracy.

## PROJECTS

01/06/2026 – 07/07/2026

### Chronic Disease Risk Predictor

- Developed an end-to-end machine learning pipeline in Python to predict the risk of chronic diseases (such as diabetes/hypertension) using patient health data.
- Processed and prepared structured health datasets by handling missing values, encoding categorical variables, and normalizing features using Pandas and NumPy.
- Trained and optimized classification models
- Implemented explainable AI techniques to interpret model outputs and identify the most critical patient risk factors, mirroring real-world clinical decision-support requirements.

01/07/2025 – 15/06/2026

### Fluorescence-Based Detection and Tracking of Trapped Particles in an Optical Tweezer System

- Constructed and aligned a modular optical tweezer setup to stably trap dielectric micro-particles.
- Integrated a fluorescence detection module to simultaneously manipulate and image the trapped specimens.
- Characterized the trap stiffness via Brownian motion analysis and utilized the fluorescence signal to study, demonstrating precision in optomechanical alignment and low-light signal detection.

26/01/2026 – 01/06/2026

### Feature Extraction for EEG Epileptic Seizure Detection and Classification

- Developed an efficient, automated machine learning system to detect and classify epileptic seizures using advanced EEG signal processing.
- Engineered robust feature extraction pipelines to prepare and process complex neurological data, optimizing the data for model training.
- Trained and evaluated classification models using Python and Scikit-learn/TensorFlow/PyTorch to replace inefficient manual detection methods.
- Deployed the model to deliver real-time, objective clinical insights into a patient's neurological status, demonstrating an ability to build AI-driven solutions for real-world healthcare challenges.

14/10/2025 – 04/04/2026

### Numerical Gradient Descent Optimizer

- Developed a numerical optimization engine in C++17 to iteratively minimize complex mathematical functions, leveraging Standard Template Library (STL) algorithms for efficient memory and data management.
- Implemented gradient-based search algorithms to perform precise linear and polynomial regression, directly applying core machine learning and probability concepts to algorithmic problem-solving.
- Engineered an object-oriented architecture utilizing inheritance and polymorphism to evaluate multiple mathematical models, incorporating robust code analysis and debugging techniques to ensure algorithmic convergence.

01/09/2025 – 30/12/2025

### Particle Tracking & Trajectory Analyzer

- Developed a high-performance 2D particle tracking application in C++.
- Engineered numerical analysis algorithms to calculate fundamental physical properties—such as velocity, displacement, and statistical trajectory variances—mirroring real-world optical trap stiffness evaluations.
- Implemented object-oriented programming (OOP) principles by designing encapsulated Particle and Trajectory classes to maintain code modularity, ensuring scalable and optimized data processing for complex simulations.

30/06/2025 – 08/08/2025

### Atomic Spectroscopy & Laser Locking

- Performed Saturated Absorption Spectroscopy (SAS) on a Cesium cell to isolate specific atomic transitions.
- Successfully locked the laser output to a hyperfine transition, demonstrating proficiency in optical alignment, feedback electronics, and laser stabilization techniques.

30/06/2025 – 08/08/2025

### Electromagnetically Induced Transparency in Cesium Vapor Cell

- Conducted an experimental study on Electromagnetically Induced Transparency (EIT) in cesium atoms to investigate quantum interference effects in a Λ-type three-level system. A diode laser tuned to the cesium D2 line was used in a probe–coupling configuration.Probe transmission was recorded using a photodetector and digital oscilloscope, enabling the observation of transparency windows in the absorption spectrum.

30/06/2025 – 08/08/2025

### Investigation of Atomic Resonance Using the Hanle Effect

- Demonstrated the Hanle effect using a compact setup with a cesium vapor cell and Helmholtz coils.
- Observed a distinct resonance peak at zero magnetic field, confirming the destruction of atomic coherence. The experiment highlights the impact of magnetic fields on quantum states that is fundamental to atomic magnetometry and precision clocks.

01/11/2024 – 16/01/2025

### Design and Simulation of Unregulated and Regulated 12V DC Power Supply

- Designed and simulated an unregulated 12V DC power supply with a 120 Ω load and a ripple of less than 1% using Proteus.
- Implemented a transformer, full-wave rectifier, and filter capacitor, and later integrated a voltage regulator for a stable output. The project enhanced my understanding of power supply design, simulation, and voltage regulation techniques.

01/02/2024 – 03/06/2024

### Statistical Analysis of Public Opinion on Online vs. Physical Classes During COVID-19

- Collected survey data via Google Forms to assess people's experiences and preferences between online and physical classes during the COVID-19 pandemic.
- Analyzed the responses using probability and statistical measures, interpreting trends and patterns in public opinion through quantitative evaluation.

## SKILLS

**Programming & AI**

Machine Learning | Deep Learning | Artificial Intelligence | Python (computer programming) | Data Science | Matplotlib | Git | TensorFlow | Scikit-learn | PyTorch | SQL | Pandas | C++ | Object oriented Programming

### Analytical & Scientific Skills

Data Visualization | Quantum Optics | Mathematical Modelling | Quantum Computing

### Tools & Software

Microsoft Word | Microsoft Office | Microsoft Excel | Microsoft Powerpoint | Web Design & Web Development | Linux

### Interpersonal & Leadership Skills

Social Media | Organizational and planning skills | Event Management | Team-work oriented | Good listener and communicator | Decision-making

## CONFERENCES AND SEMINARS

19/12/2025 – 19/12/2025 Islamabad
### Measurements an Simulations of Innovative LGAD Designs

16/12/2025 – 16/12/2026 Islamabad
### Can We Reverse Decoherence?

10/02/2025 – 12/02/2025 Islamabad
### From Chemistry to Clinics: Advances in Radiopharmaceuticals

14/10/2024 – 18/10/2024 Islamabad
### First International Conference on Recent Advances in Quantum Science and Technology

26/09/2024 – 26/09/2024 Islamabad
### From Classical Computing to Quantum Computing

29/04/2024 – 29/04/2024 Islamabad
### Lecture on Gravity

## HONOURS AND AWARDS

### Certificate of Appreciation for Social Media Head – PIEAS Society for Physics

Responsible for managing the society's online presence across platforms. Duties include creating and scheduling content, promoting events, increasing engagement, and maintaining a consistent brand voice.

### Participation Certificate – International Youth Math Challenge, IYMC

### Second Position in Space Week Short Story Competition – NUST Physics & Astronomy Society,National University of Science and Technology (NUST)

22/07/2026
### Certificate of Appreciation – Pakistan Institute of Engineering and Aplied Sciences

Certificate of Appreciation for the Best Final Year Project.

### Certificate of Appreciation for Director Rumi's Circle – PIEAS Literary Society, PIEAS

## HOBBIES AND INTERESTS

**Quantum Computing**

**Laser Cooling of Atoms**

**Machine Learning**

**Quantum Cryptography**

**Magnetic Field Sensing**

**Crocheting**

**Art and Crafts**

**Writing Poetry**

**Book Reading**

## LANGUAGE SKILLS

Mother tongue(s): **URDU** | **PANJABI; PUNJABI**

Other language(s):

| | UNDERSTANDING | | SPEAKING | | WRITING |
|---|---|---|---|---|---|
| | Listening | Reading | Spoken production | Spoken interaction | |
| **ENGLISH** | C2 | C2 | C2 | C2 | C2 |

*Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user*

## CERTIFICATIONS

Coursera
**Build Full Website Using Wordpress**

Coursera
**Create Your First Python Program From UST**

Coursera
**Search Engine Optimization (SEO) with Squarespace**